

Q1 - 24 January - Shift 1

The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is :

- (1) $\frac{25}{3}$ (2) $\frac{22}{3}$ (3) 9 (4) $\frac{23}{3}$

Space for your notes:

Q2 - 24 January - Shift 2

If the area of the region bounded by the curves $y^2 - 2y = -x$, $x + y = 0$ is A, then $8A$ is equal to

Space for your notes:

Q3 - 25 January - Shift 1

It the area enclosed by the parabolas $P_1 : 2y = 5x^2$

and $P_2 : x^2 - y + 6 = 0$ is equal to the area enclosed

by P_1 and $y = \alpha x$, $\alpha > 0$, then α^3 is equal to

Space for your notes:

Q4 - 29 January - Shift 1

Let Δ be the area of the region $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 21, y^2 \leq 4x, x \geq 1\}$. Then

$\frac{1}{2} \left(\Delta - 21 \sin^{-1} \frac{2}{\sqrt{7}} \right)$ is equal to

- (1) $2\sqrt{3} - \frac{1}{3}$ (2) $\sqrt{3} - \frac{2}{3}$
(3) $2\sqrt{3} - \frac{2}{3}$ (4) $\sqrt{3} - \frac{4}{3}$

Space for your notes:

Q5 - 29 January - Shift 1

Let $A = \{(x, y) \in \mathbb{R}^2 : y \geq 0, 2x \leq y \leq \sqrt{4 - (x-1)^2}\}$

Space for your notes:

and $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq y \leq \min\{2x, \sqrt{4 - (x-1)^2}\}\}$

Then the ratio of the area of A to the area of B is

(1) $\frac{\pi-1}{\pi+1}$

(2) $\frac{\pi}{\pi-1}$

(3) $\frac{\pi}{\pi+1}$

(4) $\frac{\pi+1}{\pi-1}$

Q6 - 29 January - Shift 2

The area of the region

Space for your notes:

$$A = \{(x, y) : |\cos x - \sin x| \leq y \leq \sin x, 0 \leq x \leq \frac{\pi}{2}\}$$

(1) $1 - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{5}}$

(2) $\sqrt{5} + 2\sqrt{2} - 4.5$

(3) $\frac{3}{\sqrt{5}} - \frac{3}{\sqrt{2}} + 1$

(4) $\sqrt{5} - 2\sqrt{2} + 1$

Q7 - 30 January - Shift 1

Let α be the area of the larger region bounded by

Space for your notes:

the curve $y^2 = 8x$ and the lines $y = x$ and $x = 2$, which lies in the first quadrant. Then the value of 3α is equal to _____.

Q8 - 30 January - Shift 2

Let q be the maximum integral value of p in $[0, 10]$ for which the roots of the equation $x^2 - px + \frac{5}{4}p = 0$ are rational. Then the area of the region $\{(x, y) : 0 \leq y \leq (x - q)^2, 0 \leq x \leq q\}$ is

- (1) 243 (2) 25
(3) $\frac{125}{3}$ (4) 164

Space for your notes:

Q9 - 30 January - Shift 2

Let A be the area of the region

$$\{(x, y) : y \geq x^2, y \geq (1 - x)^2, y \leq 2x(1 - x)\}$$

Then $540A$ is equal to

Space for your notes:

Q10 - 31 January - Shift 1

Let for $x \in \mathbb{R}$

$$f(x) = \frac{x + |x|}{2} \text{ and } g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

Then area bounded by the curve $y = (f \circ g)(x)$ and the lines $y = 0, 2y - x = 15$ is equal to _____.

Space for your notes:

Q11 - 31 January - Shift 2

Let the area of the region $\{(x, y) : |2x - 1| \leq y \leq |x^2 - x|, 0 \leq x \leq 1\}$ be A .

Then $(6A + 11)^2$ is equal to _____.

Space for your notes:

Q12 - 01 February - Shift 1

The area enclosed by the closed curve C given by

the differential equation $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$

is 4π .

Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is

(1) $2\sqrt{3}$ (2) $\frac{2\sqrt{3}}{3}$

(3) 2 (4) $\frac{4\sqrt{3}}{3}$

Space for your notes:

Q13 - 01 February - Shift 1

Let A be the area bounded by the curve $y = x|x-3|$, the x-axis and the ordinates $x = -1$ and $x = 2$. Then $12A$ is equal to _____.

Space for your notes:

Q14 - 01 February - Shift 2

If $A = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then :

(1) $A^{30} - A^{25} = 2I$

(2) $A^{30} + A^{25} + A = I$

(3) $A^{30} + A^{25} - A = I$

(4) $A^{30} = A^{25}$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)

Q2 (36)

Q3 (600)

Q4 (4)

Q5 (1)

Q6 (4)

Q7 (22)

Q8 (1)

Q9 (25)

Q10 (72)

Q11 (125)

Q12 (4)

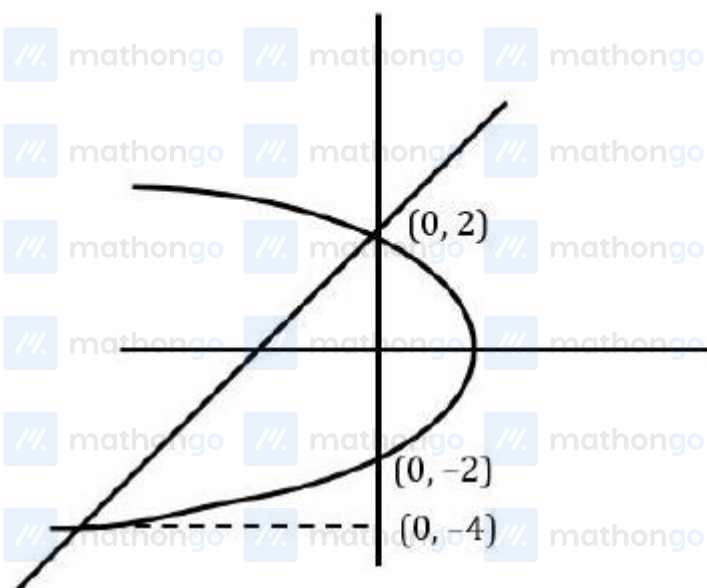
Q13 (62)

Q14 (3)

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Q1 (3)



$$y^2 + 4x = 4$$

$$y^2 = -4(x-1)$$

$$A = \int_{-4}^2 \left(\frac{4-y^2}{4} - \frac{y-2}{2} \right) dy = 9$$

Q2 (36)

$$y^2 - 2y = -x$$

$$\Rightarrow y^2 - 2y + 1 = -x + 1$$

$$(y - 1)^2 = -(x - 1)$$

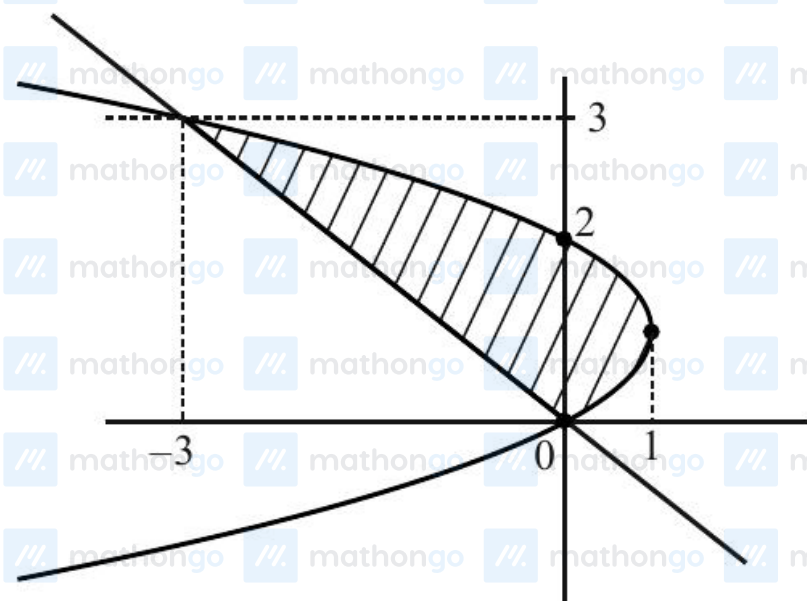
$$y = -x$$

Points of intersection

$$x^2 + 2x = -x$$

$$x^2 + 3x = 0$$

$$x = 0, -3$$



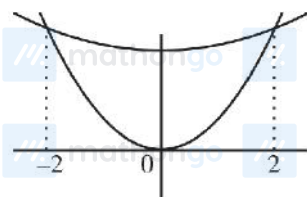
$$A = \int_0^3 (-y^2 + 2y + y) dy$$

$$= \frac{3y^2}{2} - \frac{y^3}{3} \Big|_0^3 = \frac{9}{2}$$

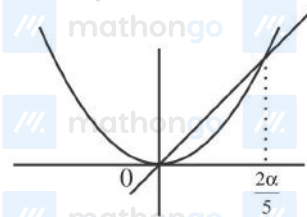
$$8A = 36$$

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Q3 (600)



Abscissa of point of intersection of $2y = 5x^2$
and $y = x^2 + 6$ is ± 2

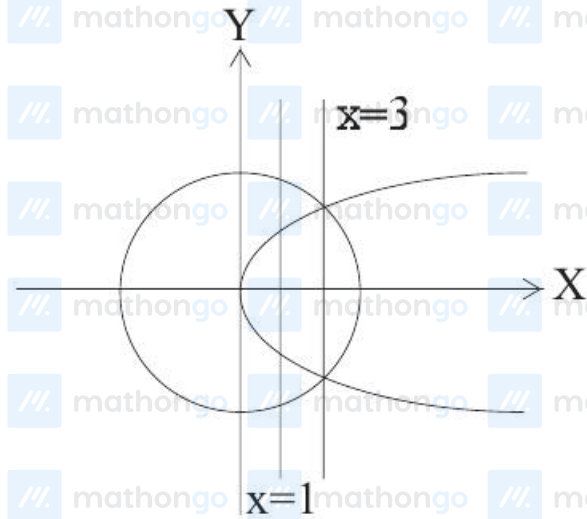


$$\text{Area} = 2 \int_0^2 \left(x^2 + 6 - \frac{5x^2}{2} \right) dx = \int_0^{\frac{2\alpha}{5}} \left(\alpha x - \frac{5x^2}{2} \right) dx$$

$$\Rightarrow \int_0^{\frac{2\alpha}{5}} \left(\alpha x - \frac{5x^2}{2} \right) dx = 16$$

$$\Rightarrow \alpha^3 = 600$$

Q4 (4)



$$\text{Area} = 2 \int_1^3 2\sqrt{x} \, dx + 2 \int_3^{\sqrt{21}} \sqrt{21-x^2} \, dx$$

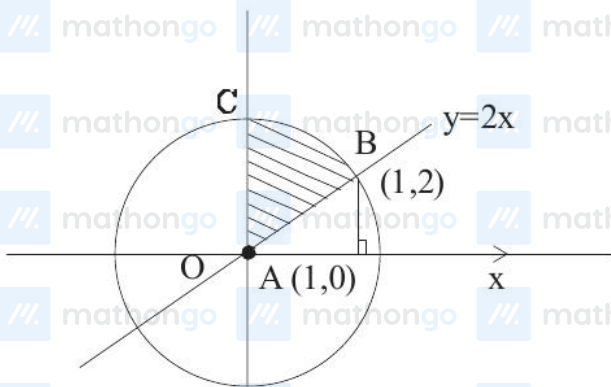
$$\Delta = \frac{8}{3}(3\sqrt{3}-1) + 21 \sin^{-1}\left(\frac{2}{\sqrt{7}}\right) - 6\sqrt{3}$$

$$\frac{1}{2} \left(\Delta - 21 \sin^{-1}\left(\frac{2}{\sqrt{7}}\right) \right) = \frac{2\sqrt{3} - \frac{8}{3}}{2}$$

$$= \sqrt{3} - \frac{4}{3}$$

Q5 (1)

$$y^2 + (x-1)^2 = 4$$

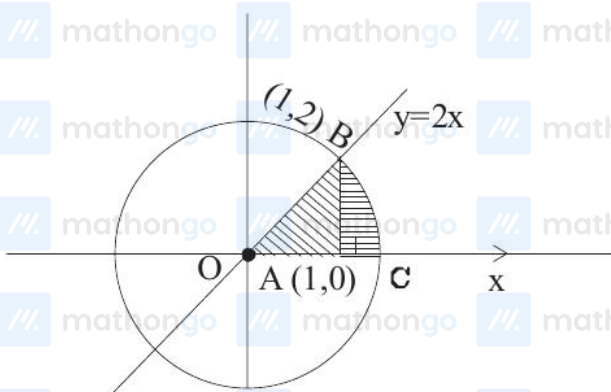


shaded portion = circular (OABC)

$$- \text{Ar}(\triangle OAB)$$

$$= \frac{\pi(4)}{4} - \frac{1}{2}(2)(1)$$

$$A = (\pi - 1)$$



Area B = Ar ($\triangle AOB$) + Area of arc of circle (ABC)

$$= \frac{1}{2}(1)(2) + \frac{\pi(2)^2}{4} = \pi + 1$$

$$\frac{A}{B} = \frac{\pi - 1}{\pi + 1}$$

Q6 (4)

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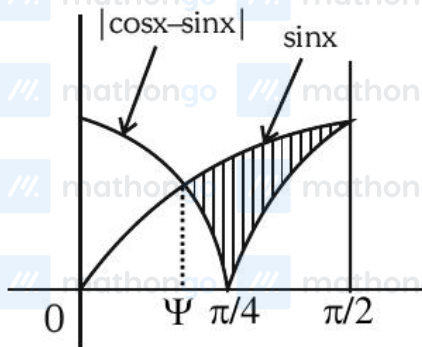
$$|\cos x - \sin x| \leq y \leq \sin x$$

Intersection point of $\cos x - \sin x = \sin x$

$$\Rightarrow \tan x = \frac{1}{2}$$

$$\text{Let } \psi = \tan^{-1} \frac{1}{2}$$

$$\text{So, } \tan \psi = \frac{1}{2}, \sin \psi = \frac{1}{\sqrt{5}}, \cos \psi = \frac{2}{\sqrt{5}}$$



$$\text{Area} = \int_{\psi}^{\pi/2} (\sin x - |\cos x - \sin x|) dx$$

$$= \int_{\psi}^{\pi/4} (\sin x - (\cos x - \sin x)) dx$$

$$+ \int_{\pi/4}^{\pi/2} (\sin x - (\sin x - \cos x)) dx$$

$$= \int_{\psi}^{\pi/4} (2 \sin x - \cos x) dx + \int_{\pi/4}^{\pi/2} \cos x dx$$

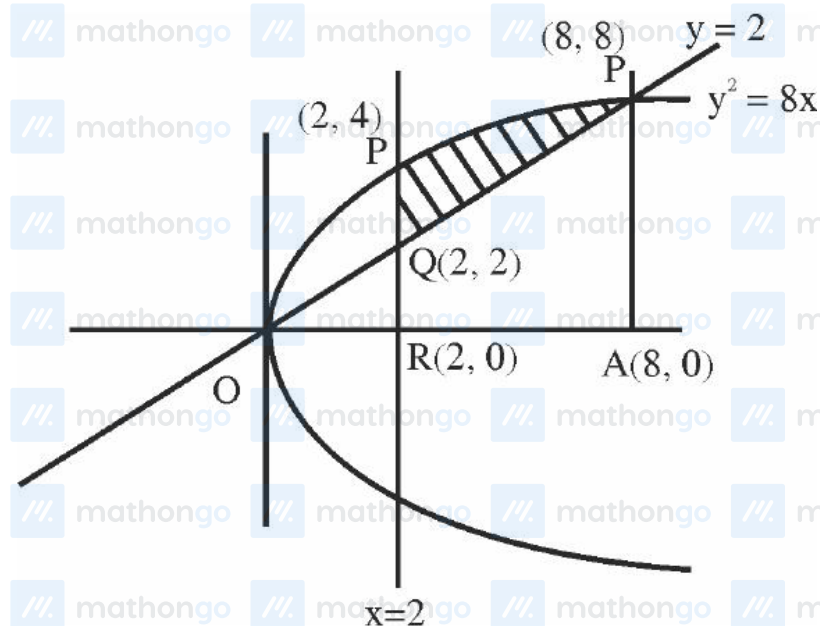
$$= [-2 \cos x - \sin x]_{\psi}^{\pi/4} + [\sin x]_{\pi/4}^{\pi/2}$$

$$= -\sqrt{2} - \frac{1}{\sqrt{2}} + 2 \cos \psi + \sin \psi + \left(1 - \frac{1}{\sqrt{2}}\right)$$

$$= -\sqrt{2} - \frac{1}{\sqrt{2}} + 2 \left(\frac{2}{\sqrt{5}}\right) + \left(\frac{1}{\sqrt{5}}\right) + 1 - \frac{1}{\sqrt{2}}$$

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$$y = x$$

$$\& y^2 = 8x$$

Solving it

$$x^2 = 8x$$

$$\therefore x = 0, 8$$

$$\therefore y = 0, 8$$

$x = 2$ will intersect occur at

$$y^2 = 16 \Rightarrow y = \pm 4$$

$\therefore$ Area of shaded

$$= \int_2^8 (\sqrt{8x} - x) dx = \int_2^8 (2\sqrt{2}\sqrt{x} - x) dx$$

$$= \left[2\sqrt{2} \cdot \frac{x^{3/2}}{3/2} - \frac{x^2}{2} \right]_2^8$$

$$= \left(\frac{4\sqrt{2}}{3} \cdot 2^{9/2} - 32 \right) - \left(\frac{4\sqrt{2}}{3} \cdot 2^{9/2} - 2 \right)$$

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$$\therefore 3A = 22$$

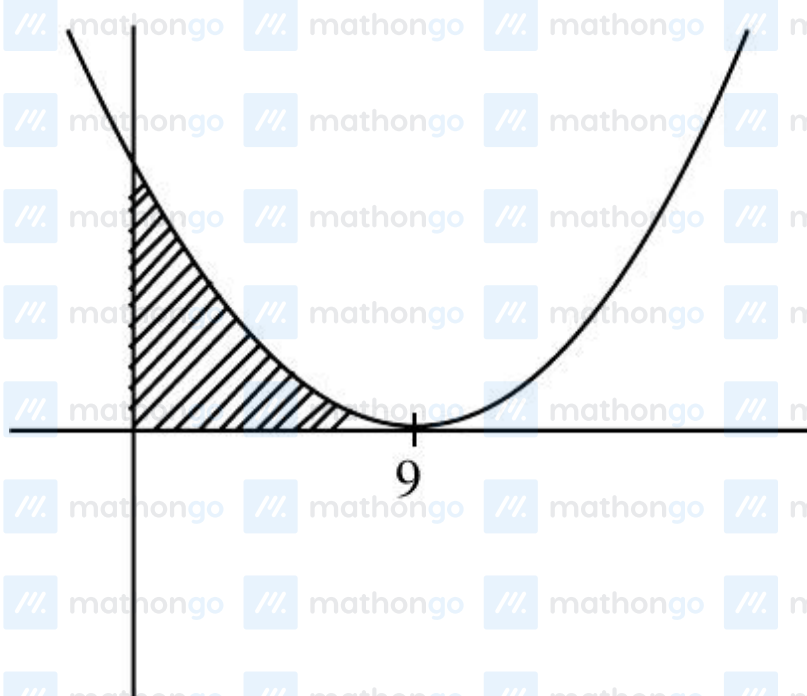
Q8 (1)

$$x^2 - px + \frac{5p}{4} = 0$$

$$D = p^2 - 5p = p(p - 5)$$

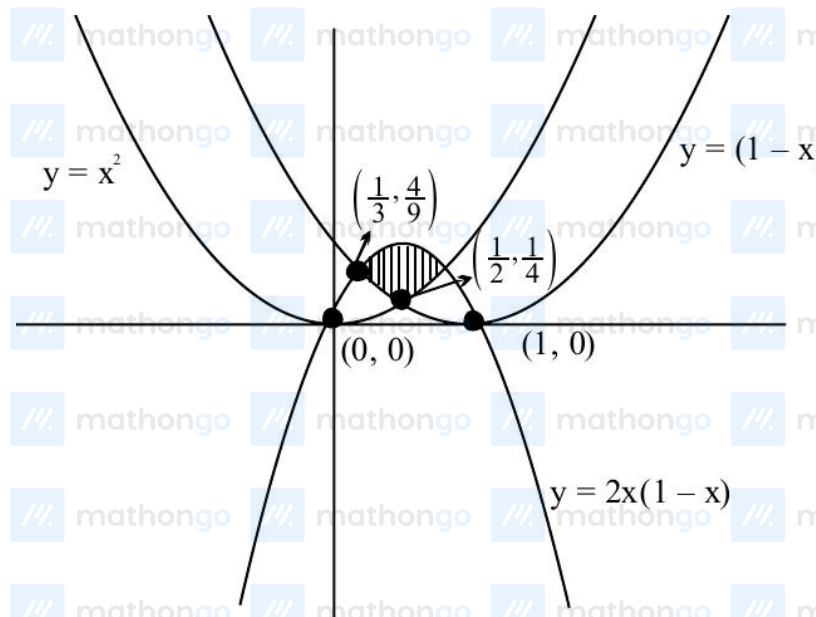
$$\therefore q = 9$$

$$0 \leq y \leq (x - 9)^2$$



$$\text{Area} = \int_0^9 (x - 9)^2 dx = 243$$

Q9 (25)



$$\begin{aligned}
 A &= 2 \int_{\frac{1}{3}}^{\frac{1}{2}} (2x - 2x^2 - (1-x)^2) dx \\
 &= 2 \left[2x^2 - x^3 - x \right]_{1/3}^{1/2} \\
 \therefore A &= \frac{5}{108} \Rightarrow 540A = \frac{5}{108} \times 540 = 25
 \end{aligned}$$

Q10 (72)

$$f(x) = \frac{x + |x|}{2} = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 & x \geq 0 \\ x & x < 0 \end{cases}$$

$$f \circ g(x) = f[g(x)] = \begin{cases} g(x) & g(x) \geq 0 \\ 0 & g(x) < 0 \end{cases}$$

$$f \circ g(x) = \begin{cases} x^2 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

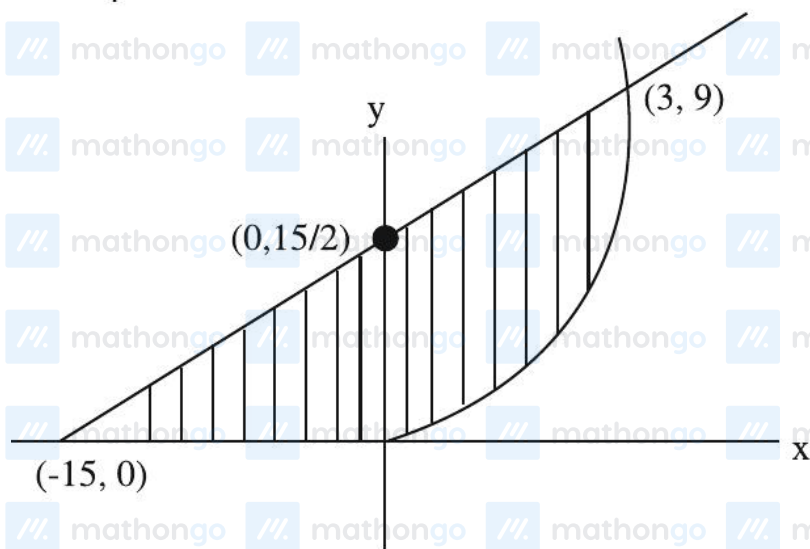
$$2y - x = 15$$

$$A = \int_0^3 \left(\frac{x+15}{2} - x^2 \right) dx + \frac{1}{2} \times \frac{15}{2} \times 15$$

$$\left. \frac{x^2}{4} + \frac{15x}{2} - \frac{x^3}{3} \right|_0^3 + \frac{225}{4}$$

$$= \frac{9}{4} + \frac{45}{2} - 9 + \frac{225}{4} = \frac{99 - 36 + 225}{4}$$

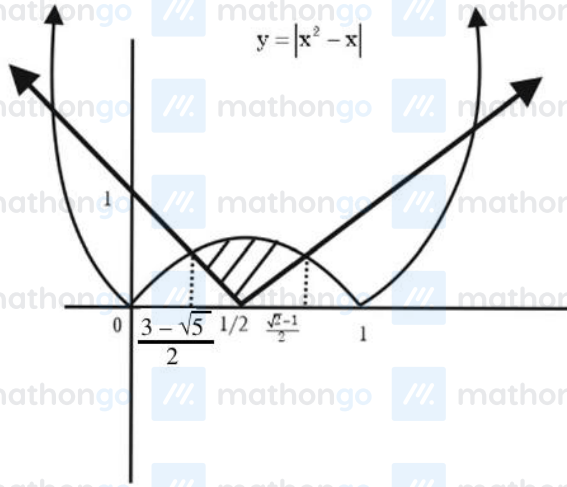
$$= \frac{288}{4} = 72$$



Q11 (125)

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$$y \geq |2x - 1|, y \leq |x^2 - x|$$



Both curve are symmetric about $x = \frac{1}{2}$ Hence

$$A = 2 \int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} ((x-x^2) - (1-2x)) dx$$

$$A = 2 \int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} (-x^2 + 3x - 1) dx = 2 \left(\frac{-x^3}{3} + \frac{3}{2}x^2 - x \right) \Big|_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}}$$

On solving $6A + 11 = 5\sqrt{5}$

Q12 (4)

$$\frac{dy}{dx} + \frac{x+a}{y-2} = 0$$

$$\frac{dy}{dx} = \frac{x+a}{2-y}$$

$$(2-y) dy = (x+a) dx$$

$$2y \frac{-y}{2} = \frac{x^2}{2} + ax + c$$

$$a + c = -\frac{1}{2} \text{ as } y(1) = 0$$

$$X^2 + y^2 + 2ax - 4y - 1 - 2a = 0$$

$$\pi r^2 = 4\pi$$

$$r^2 = 4$$

$$4 = \sqrt{a^2 + 4 + 1 + 2a}$$

$$(a+1)^2 = 0$$

$$P, Q = (0, 2 \pm \sqrt{3})$$

$$\text{Equation of normal at P, Q are } y - 2 = \sqrt{3}(x - 1)$$

$$y - 2 = -\sqrt{3}(x - 1)$$

$$R = \left(1 - \frac{2}{\sqrt{3}}, 0\right)$$

$$S = \left(1 + \frac{2}{\sqrt{3}}, 0\right)$$

$$RS = \frac{4}{\sqrt{3}} = 4 \frac{\sqrt{3}}{3}$$

Q13 (62)

$$A = \int_{-1}^0 (x^2 - 3x) dx + \int_0^2 (3x - x^2) dx$$

$$\Rightarrow A = \left. \frac{x^3}{3} - \frac{3x^2}{2} \right|_{-1}^0 + \left. \frac{3x^2}{2} - \frac{x^3}{3} \right|_0^2$$

$$\Rightarrow A = \frac{11}{6} + \frac{10}{3} = \frac{31}{6}$$

$$\therefore 12A = 62$$

Q14 (3)

$$A = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{bmatrix}$$

$$\text{If } A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \text{ Here } \alpha = \frac{\pi}{3}$$

$$A^2 = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$A^{30} = \begin{bmatrix} \cos 30\alpha & \sin 30\alpha \\ -\sin 30\alpha & \cos 30\alpha \end{bmatrix}$$

$$A^{30} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^{25} = \begin{bmatrix} \cos 25\alpha & \sin 25\alpha \\ -\sin 25\alpha & \cos 25\alpha \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

$$A^{25} = A$$

$$A^{25} - A = 0$$